

Hospital Management System — Project Report

Name: Sanya Sinha

Roll no:24F1002663

Course:BS in data Science

Project Title: Hospital Management System

TechnologiesUsed: Flask, SQLAlchemy, SQLite, HTML, CSS, Jinja

1. Introduction

The Hospital Management System is a web-based application built to streamline hospital operations.

It provides a secure platform for three types of users:

Admin – the supreme leader-manages hospital staff and appointments.

Doctor – manages appointments, updates treatment and availability

Patient – books appointments, edits profile, and views history

The goal of this project is to design a simple, functional, and efficient system using Python and Flask.

2. Objectives

The system was designed to:

- Provide role-based access for Admin, Doctor, and Patient
- Allow patients to book or cancel appointments
- Allow doctors to mark appointments, provide treatment, and maintain history
- Enable admin to manage doctors, patients, and appointments
- Prevent double-booking and restrict bookings on unavailable days
- Maintain detailed treatment history
- Blacklist user and prevent them from logging
- Emails the credentials of the newly registered doctor automatically

3. Technology Stack

Backend

Flask (Python) – handles routing and backend logic

SQLAlchemy ORM – models & database interaction

SQLite – lightweight relational database

Frontend

HTML5 & CSS3

Jinja templating engine

Security

Password hashing using **Werkzeug**

Session-based authentication

Role-based access system

4. System Modules

A. Admin Module

- Login with admin credentials
- View total doctors, patients, and appointments
- Register new doctors (with automated email credentials)
- Search and view doctor profiles
- Update departments & specializations
- Search and view patients
- Blacklist / Unblacklist users
- Add, edit, or delete appointments
- Prevent double booking
- Prevent booking when doctor is unavailable

B. Doctor Module

- View weekly upcoming appointments
- View past appointments
- Mark appointments as Completed / Cancelled
- Add treatment notes and prescriptions
- View full medical history of any assigned patient
- Manage weekly availability schedule
- Edit personal profile

C. Patient Module

- Register & login
- Search doctors by name or specialization
- View doctor weekly availability
- Book appointments
- Prevent booking unavailable days or already-booked time slots
- Cancel appointments
- View upcoming & past appointments
- View treatment history
- Edit profile

5. Database Models

User: id, name, email, password, contact, role, status, is_blacklisted

Doctor: id, specialization, **departmentid**, is_available

Patient: id, dob, gender

Department: id, name, description

Appointment: id, **patientid**, **doctorid**, date, time, status

Treatment: id, **patientid**, **doctorid**, diagnosis, procedure, date

Prescription: id, **treatmentid**, medication

DoctorAvailability: id, **doctor_id**, date, is_available

Relationships are defined using SQLAlchemy ORM.

6. Application Flow

Admin Flow

Login → Dashboard → Register Doctor → Manage Doctors → Manage Patients → Manage Appointments

Doctor Flow

Login → Dashboard → Mark appointment → Enter treatment → View history → Update availability

Patient Flow

Register/Login → Search Doctor → Check Availability → Book Appointment → View History → Edit Profile

7. Testing

Appointment conflicts tested

Availability checked before booking

Login tested with all roles

CRUD operations tested for all modules

Treatment and prescription records validated

8. Conclusion

The Hospital Management System meets all required functionalities and successfully manages the workflow between Admin, Doctors, and Patients. The project is scalable, cleanly structured, and easy to maintain.

I hereby declare that this project is my own work.

AI tools (ChatGPT, GitHub Copilot, etc.) were used only for:

- Code debugging support: **Yes**
- UI/HTML template help: **Yes**

AI Usage %: 10–20%

I understand that the core logic, design, and implementation were done by me.

Link to the Video->

https://drive.google.com/file/d/1Z8d7x3b8CWQY2PP_mOCdG1G2B5x3lUdX/view?usp=sharing